

Classification of Network Traffic (BENIGN, LDAP, NetBIOS) using Machine Learning

Objective and Approach

The goal of this analysis was to identify the type of network traffic and classify it into one of three categories: BENIGN, LDAP, or NetBIOS.

BENIGN traffic refers to normal and legitimate network activity such as regular communication between systems. **LDAP** traffic is related to directory services used for authentication, but in this dataset it represents LDAP-based attack traffic. **NetBIOS** traffic is generally used in Windows systems for communication and resource sharing, but here it represents NetBIOS-based attack traffic.

So overall, the main aim was to differentiate between normal traffic and attack traffic based on how the network behaves.

Feature Selection

The original dataset contained more than 80 columns. But using all of them would not really help and would make the model more complex. So the dataset was reduced to around 20 important features.

These features were selected based on what they represent:

- Traffic Volume (how much data is sent)
- Traffic Rate (how fast data is sent)
- Packet Structure (size and shape of packets)
- Connection Behaviour (SYN, ACK, etc.)
- Directional Features (flow between source and destination)
- Timing Features (how traffic behaves over time)

These were chosen because they actually describe how the traffic behaves, not just raw numbers.

Type of Analysis

The analysis performed here is Supervised Machine Learning using a Multi-Class Classification approach.

In simple terms, the model is trained using labelled data and it learns patterns to classify traffic into different categories.

Methodology

The work was done step by step.

First, the dataset was cleaned by removing unnecessary columns and handling missing values. Then feature selection was done to keep only useful columns.

```
===== RESTART: D:/Priyanka Keshava/Programming/Python/ABC/Cleaning.py =====
☒ Dataset loaded

Columns after cleaning:
['Unnamed: 0', 'Flow ID', 'Source IP', 'Source Port', 'Destination IP', 'Destination Port', 'Protocol', 'Timestamp', 'Flow Duration', 'Total Fwd Packets', 'Total Backward Packets', 'Total Length of Fwd Packets', 'Total Length of Bwd Packets', 'Fwd Packet Length Max', 'Fwd Packet Length Min', 'Fwd Packet Length Mean', 'Fwd Packet Length Std', 'Bwd Packet Length Max', 'Bwd Packet Length Min', 'Bwd Packet Length Mean', 'Bwd Packet Length Std', 'Flow Bytes/s', 'Flow Packets/s', 'Flow IAT Mean', 'Flow IAT Std', 'Flow IAT Max', 'Flow IAT Min', 'Fwd IAT Total', 'Fwd IAT Mean', 'Fwd IAT Std', 'Fwd IAT Max', 'Fwd IAT Min', 'Bwd IAT Total', 'Bwd IAT Mean', 'Bwd IAT Std', 'Bwd IAT Max', 'Bwd IAT Min', 'Fwd PSH Flags', 'Bwd PSH Flags', 'Fwd URG Flags', 'Bwd URG Flags', 'Fwd Header Length', 'Bwd Header Length', 'Fwd Packets/s', 'Bwd Packets/s', 'Min Packet Length', 'Max Packet Length', 'Packet Length Mean', 'Packet Length Std', 'Packet Length Variance', 'FIN Flag Count', 'SYN Flag Count', 'RST Flag Count', 'PSH Flag Count', 'ACK Flag Count', 'URG Flag Count', 'CWE Flag Count', 'ECE Flag Count', 'Down/Up Ratio', 'Average Packet Size', 'Avg Fwd Segment Size', 'Avg Bwd Segment Size', 'Fwd Header Length.1', 'Fwd Avg Bytes/Bulk', 'Fwd Avg Packets/Bulk', 'Fwd Avg Bulk Rate', 'Bwd Avg Bytes/Bulk', 'Bwd Avg Packets/Bulk', 'Bwd Avg Bulk Rate', 'Subflow Fwd Packets', 'Subflow Fwd Bytes', 'Subflow Bwd Packets', 'Subflow Bwd Bytes', 'Init_Win_bytes_forward', 'Init_Win_bytes_backward', 'act_data_pkt_fwd', 'min_seg_size_forward', 'Active Mean', 'Active Std', 'Active Max', 'Active Min', 'Idle Mean', 'Idle Std', 'Idle Max', 'Idle Min', 'SimilarHTTP', 'Inbound', 'Label']

☒ After cleaning NaN/inf: (2059122, 82)

☒ Selected features

Label distribution BEFORE sampling:
Label
LDAP      1862215
NetBIOS    191854
BENIGN     5053
Name: count, dtype: int64

☒ CLEANED DATASET SAVED AS: cleaned_ddos_data.csv

Final dataset shape: (45053, 21)

Final label distribution:
Label
NetBIOS    20000
LDAP        20000
BENIGN      5053
Name: count, dtype: int64
```

After that, the dataset was balanced because initially one class had much more data than the others.

Then a Random Forest model was trained. This model works by using multiple decision trees and combining their results.

Finally, the model was evaluated using accuracy, confusion matrix and classification report.

```
===== RESTART: D:/Priyanka Keshava/Programming/Python/ABC/Model.py =====
☒ Dataset loaded: (45053, 21)

✧ Label Mapping:
BENIGN -> 0
LDAP -> 1
NetBIOS -> 2

Training Size: (36042, 20)
Testing Size: (9011, 20)

☒ Model trained successfully

📊 Accuracy: 0.9996670735767396

📊 Confusion Matrix:
[[1010   0   1]
 [  0 4000   0]
 [  2   0 3998]]

📊 Classification Report:
              precision    recall  f1-score   support

    BENIGN       1.00      1.00      1.00     1011
      LDAP       1.00      1.00      1.00     4000
    NetBIOS       1.00      1.00      1.00     4000

   accuracy       1.00      1.00      1.00     9011
  macro avg       1.00      1.00      1.00     9011
weighted avg       1.00      1.00      1.00     9011

☒ Model saved as ddos_model.pkl

☒ All visualizations generated successfully.
```

Results and Analysis

After training, the model achieved an accuracy of 99.97% approx. This value initially seemed very high. However, after analyzing the data and the graphs, it became clear because the dataset has very distinct patterns between the classes.

The box plot showed that packet sizes for LDAP, NetBIOS, and BENIGN traffic are clearly separated, and the feature importance graph also indicated that packet-related features dominate the model's decisions.

Because of this clear separation, the model is able to classify the data very easily, which explains the high accuracy.

The confusion matrix further showed that almost all predictions were moreover correct and only very few errors were found.

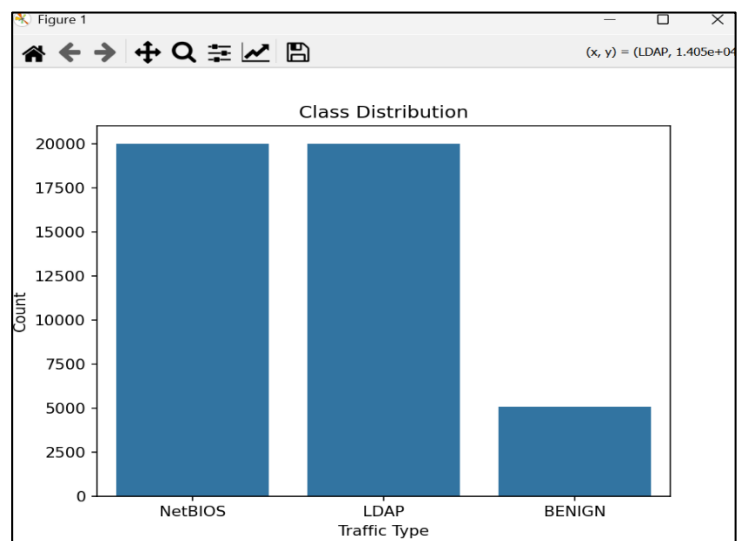
The classification report also showed that precision, recall and F1-score values were close to 1 for all classes, showing that the model performs consistently across all categories.

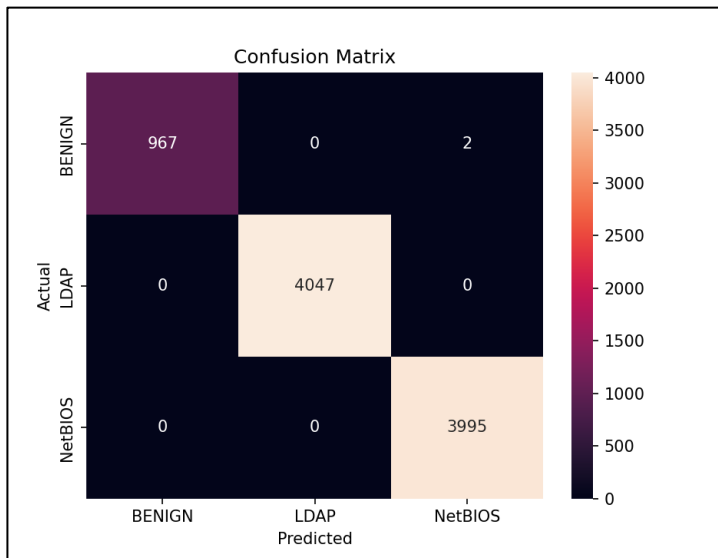
Graph Analysis

1. Class Distribution Graph

This graph shows how many samples are present for each traffic type. It was observed that LDAP and NetBIOS had much higher values compared to BENIGN traffic. The dataset was heavily imbalanced, especially because BENIGN traffic had very limited samples (around 5000) compared to the others.

Because of this, sampling was applied to reduce the size of LDAP and NetBIOS data, while keeping all available BENIGN samples. This helped in making the dataset more manageable for training, even though it was not perfectly balanced.





2. Confusion Matrix

The confusion matrix shows how well the model predicted the results.

Most of the values were on the diagonal, which means the predictions were correct. Only a few values were outside the diagonal, showing very small errors.

So basically, the model was able to classify almost all data correctly.

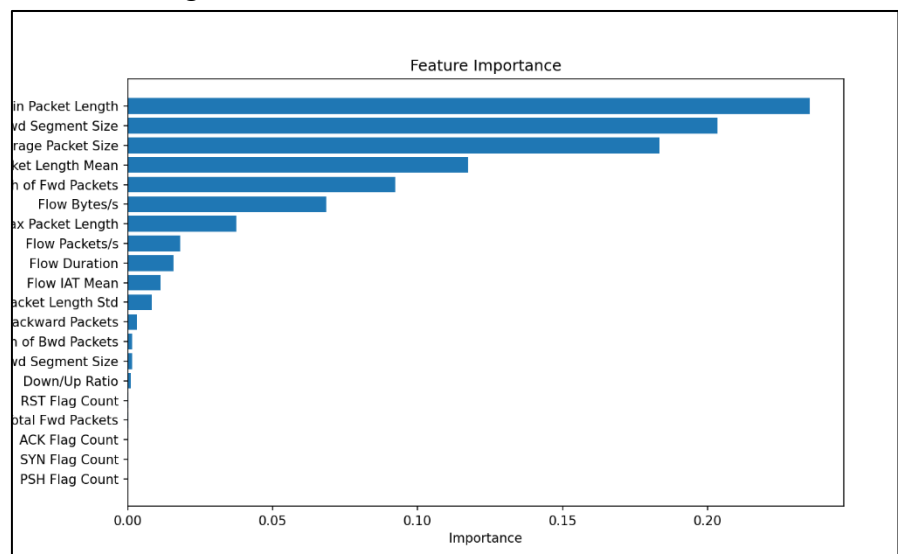
3. Feature Importance Graph

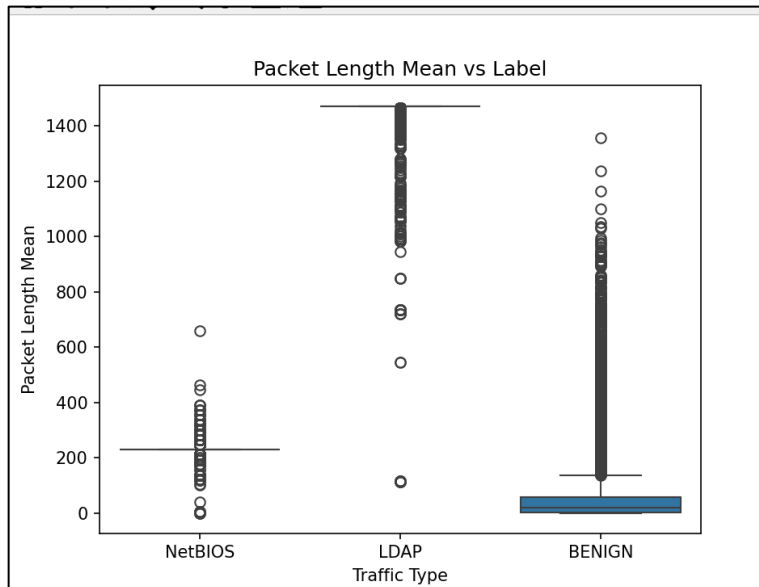
This graph shows which features the model is using the most.

It was observed that packet size-related features such as Packet Length Mean, Average Packet Size, and Segment Size had the highest importance.

This means the model mainly depends on packet size to make decisions.

At the same time, features like SYN, ACK, PSH flags had very low importance, meaning they did not contribute much.





4. Box Plot (Packet Length Mean vs Label)

This graph shows how packet size varies for each type of traffic. At first it may look like BENIGN has higher values because of some points going very high, but those are only a few extreme values (outliers). Most of the BENIGN values are actually concentrated in the lower range. LDAP traffic shows consistently high packet sizes, while NetBIOS lies in the medium range. The main part of the data (the box region) for each class is clearly

separated, even if a few values overlap.

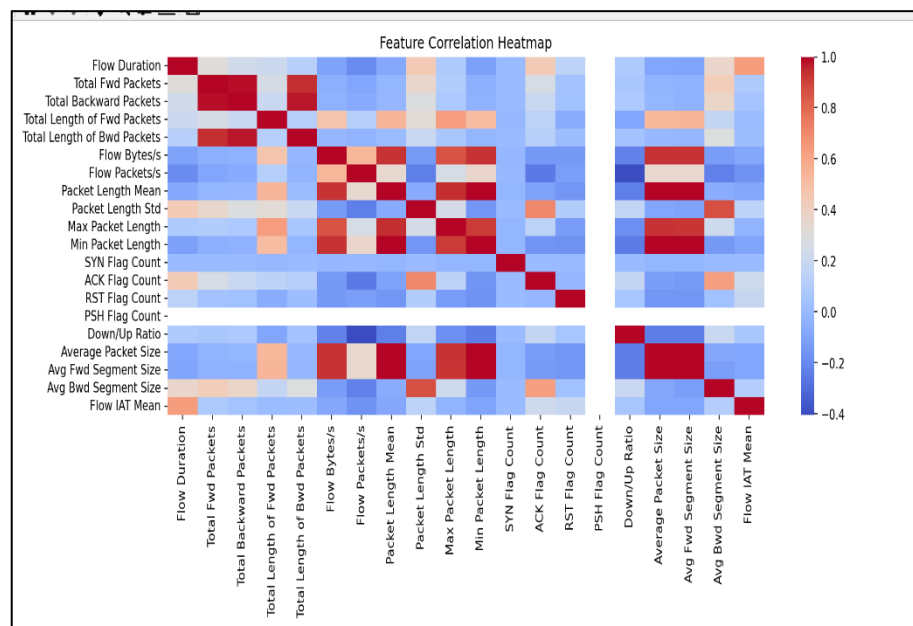
Because most of the values are different for each traffic type, the model is able to classify them easily, which explains the high accuracy.

5. Correlation Heatmap

This graph shows how different features are related to each other. The colours represent the strength of the relationship between two features. Red indicates a strong positive relationship, blue indicates a negative relationship, and lighter colours like white or peach show weak or no strong relationship.

This graph was mainly used to understand whether multiple features are giving the same type of information. It was observed that many packet-related features are highly correlated, which means they are very similar in nature.

Because of this, the graph helps in identifying redundant features. If many features are strongly related, keeping all of them is not necessary. This also explains why removing some features did not affect the model much, since similar information is still available through other features.



Interpretation of Results

To understand this better, different graphs had been used.

The class distribution graph helped to visualize that how the data is spread and it also showed that the dataset was highly imbalanced because BENIGN traffic had very less samples compared to the others.

The confusion matrix checked how the model is performing and it showed that almost all predictions were correct having only very few errors.

The feature importance graph helped to understand which features the model is dominant and it showed that packet-related features have the highest importance.

The box plot was used to compare values across different traffic types and it showed that packet sizes are clearly different for each class, which makes classification easier.

The correlation heatmap was used to check how features are related to each other and it showed that many features are highly correlated and give similar information.

Overall, these graphs helped to understanding not only the performance of the model and visualize the dataset.

Conclusion

From this analysis, it can be seen that the model is performing very well mainly because the data itself is quite easy to separate.

One thing that clearly stood out was that packet size is playing a major role, and from the graphs it is clearly different for LDAP, NetBIOS and BENIGN traffic. Because of this, the model is not really struggling and is able to classify the data easily.

It was also noticed that many features are giving similar information, so even if some features are removed, it does not make much difference in the performance.

Another observation was that the dataset is highly imbalanced, as BENIGN traffic had very less samples compared to LDAP and NetBIOS, which shows that attack traffic is more dominant in the dataset.

Also, even though the dataset is based on real attack types, it is generated in a controlled environment, so the data is quite clean and structured. Because of this, the patterns are clearer and the model is able to learn them easily. In real-world scenarios, where the data is noisier the performance may not be exactly the same.

Overall, the model works very well, but the results are also influenced by how the dataset is designed and structured.

